

torch.linalg.matrix_power#

https://pytorch.org/docs/stable/generated/torch.linalg.matrix_power.html

Description:

Computes the n -th power of a square matrix for an integer n . Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. If $n = 0$, it returns the identity matrix (or batch) of the same shape as A . If n is negative, it returns the inverse of each matrix (if invertible) raised to the power of $\text{abs}(n)$. Consider using `torch.linalg.solve()` if possible for multiplying a matrix on the left by a negative power as, if $n > 0$:

Function Signature

```
torch.linalg.matrix_power(A, n, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Raises

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
torch.linalg.solve(matrix_power(A, n), B) == matrix_power(A, -n)
@ B
```

```
>>> A = torch.randn(3, 3)
>>> torch.linalg.matrix_power(A, 0)
tensor([[1., 0., 0.],
        [0., 1., 0.],
        [0., 0., 1.]])
>>> torch.linalg.matrix_power(A, 3)
tensor([[ 1.0756,  0.4980,  0.0100],
        [-1.6617,  1.4994, -1.9980],
        [-0.4509,  0.2731,  0.8001]])
>>> torch.linalg.matrix_power(A.expand(2, -1, -1), -2)
tensor([[[ 0.2640,  0.4571, -0.5511],
         [-1.0163,  0.3491, -1.5292],
         [-0.4899,  0.0822,  0.2773]],
        [[ 0.2640,  0.4571, -0.5511],
         [-1.0163,  0.3491, -1.5292],
         [-0.4899,  0.0822,  0.2773]]])
```

Notes & Warnings

NOTE

Note Consider using `torch.linalg.solve()` if possible for multiplying a matrix on the left by a negative power as, if $n > 0$:
`torch.linalg.solve(matrix_power(A, n), B) == matrix_power(A, -n) @ B` It is always preferred to use `solve()` when possible, as it is faster and more numerically stable than computing $A^{-n}A^{-n}A^{-n}$ explicitly.

NOTE

See also `torch.linalg.solve()` computes `A.inverse() @ B` with a numerically stable algorithm.

Detailed Documentation

Documentation

Computes the n -th power of a square matrix for an integer n .

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

If $n = 0$, it returns the identity matrix (or batch) of the same shape as A . If n is negative, it returns the inverse of each matrix (if invertible) raised to the power of $\text{abs}(n)$.

Consider using `torch.linalg.solve()` if possible for multiplying a matrix on the left by a negative power as, if $n > 0$:

It is always preferred to use `solve()` when possible, as it is faster and more numerically stable than computing $A^{-n}A^{\{-n\}}A^{-n}$ explicitly.

`torch.linalg.solve()` computes $A.\text{inverse()} @ B$ with a numerically stable algorithm.

A (Tensor) – tensor of shape $(*, m, m)$ where $*$ is zero or more batch dimensions.

n (int) – the exponent.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

RuntimeError – if $n < 0$ and the matrix A or any matrix in the batch of matrices A is not invertible.
